D3 Decision Workshop

Cross-Validation Tolerance Definition — Phase 1 Pre-Build

Project	*Tungsten Supply Chain Compliance Platform*	**Doc ref**	D3-WORKSHOP-v1.0
Decision	D3 — Cross-validation tolerances	**Date**	
Status	**OPEN — required before Phase 1 build**	**Version**	1.0 DRAFT

> ⚠ *This document must be completed and signed off before Phase 1 Compliance Rule Engine build begins. Per the platform specification: "Development discretion shall not be used as a substitute."*

1. Purpose

Decision D3 requires the Compliance Lead and Supply Chain Expert to define numeric cross-validation tolerances for the Compliance Rule Engine. These values will be stored as a versioned configuration artefact and applied deterministically at runtime to flag anomalies in custody event data.

Three tolerance types must be defined per applicable event type:

Quantity tolerance (%) — Maximum acceptable percentage difference between reported input and output quantities across a processing or transfer event.

Date tolerance (days) — Maximum number of days between a linked pair of events before the gap is flagged for review.

Mass balance threshold (%) — Maximum acceptable percentage by which output mass may exceed input mass across a processing event.

2. Industry Benchmark Reference

The following standards and sources should inform the workshop discussion. Agreed tolerance values should be traceable to at least one of these references or to documented pilot supplier data.

Standard / Source	Applies to	Typical Value	Notes
ASTM E1108 / ISO 13909	Sampling / weighing	±0.5 – 1.0%	Mineral concentrates — quantity tolerance benchmark
RMAP Smelter Audit Reports	Smelting (Event 5)	85 – 92% yield	APT recovery from concentrate. Per-smelter figures available.
OECD DDG Annex II	All event types	No fixed value	Sets due diligence obligation — platform thresholds must be defensible against this standard
CBP ACE Filing Rules	Events 8 & 10	±2% quantity	US Customs filing tolerance for mineral shipments
Pilot Supplier Data	All applicable	TBD	Validate proposed tolerances against real supplier batch records before finalising

3. Per-Event Workshop Guidance

Review each event type in the workshop. The guidance column identifies the primary benchmark source and flags where tighter scrutiny is warranted.

#	Event Type	Guidance for workshop discussion
1	**Mine Extraction**	Weighbridge variance typical 0.5–1%. No linked prior event — date tolerance N/A.
2	**Concentration / Beneficiation**	Mass balance critical — tungsten recovery from ore typically 60–80%. Date spans processing range.
3	**Trading / Ownership Transfer**	Legal vs physical transfer may differ by days. Quantity should match prior custody event exactly.
4	**Laboratory Assay**	Quantity tolerance N/A (sample, not full batch). Date tolerance = time from extraction to assay receipt.
5	**Primary Processing (Smelting)**	APT yield from concentrate ~85–92%. RMAP smelter audit reports are primary benchmark source.
6	**Secondary Processing (Conversion)**	Conversion yield varies by product (oxide, carbide). Facility's own documented recovery rate is the reference.
7	**Warehousing / Storage**	Storage quantity should match inbound exactly. Date tolerance = max acceptable delay between receipt and logging.
8	**Export Clearance**	CBP ACE cross-validation applies. Date tolerance = gap between physical departure and declaration date.
9	**In-Transit Monitoring**	Quantity not re-weighed in transit — tolerance N/A. Date tolerance = waypoint timestamp lag.
10	**Import Clearance**	Section 232 tariff quantities must match export declaration. Date = port arrival to declaration filing.
11	**Customer Delivery**	Final signed receipt quantity vs shipment manifest. Date = dispatch to signed receipt.
12	**End-of-Life / Recycling Re-entry**	HIGH INTEGRITY event. Provenance break acknowledged. Tighter tolerances recommended.

4. Agreed Tolerance Values — TO BE COMPLETED

Complete all applicable cells. For each value entered, record the reference standard or data source used to justify it. N/A cells are pre-populated where the tolerance type is not applicable to that event type.

> ⚠ *Values marked N/A have been excluded on the basis that the relevant metric is not measured at that event stage. If your supply chain differs, override N/A and document the reason.*

#	Event Type	Qty Tol (%)	Date Tol (days)	Mass Bal (%)	Reference / Source
1	Mine Extraction	*Enter %*	N/A	N/A	*Cite standard/data*
2	Concentration / Beneficiation	*Enter %*	*Enter days*	*Enter %*	*Cite standard/data*
3	Trading / Ownership Transfer	*Enter %*	*Enter days*	N/A	*Cite standard/data*
4	Laboratory Assay	N/A	*Enter days*	N/A	*Cite standard/data*

#	Event Type	Qty Tol (%)	Date Tol (days)	Mass Bal (%)	Reference / Source
5	Primary Processing (Smelting)	*Enter %*	*Enter days*	*Enter %*	*Cite standard/data*
6	Secondary Processing (Conversion)	*Enter %*	*Enter days*	*Enter %*	*Cite standard/data*
7	Warehousing / Storage	*Enter %*	*Enter days*	N/A	*Cite standard/data*
8	Export Clearance	*Enter %*	*Enter days*	N/A	*Cite standard/data*
9	In-Transit Monitoring	N/A	*Enter days*	N/A	*Cite standard/data*
10	Import Clearance	*Enter %*	*Enter days*	N/A	*Cite standard/data*
11	Customer Delivery	*Enter %*	*Enter days*	N/A	*Cite standard/data*
12	End-of-Life / Recycling Re-entry	*Enter %*	*Enter days*	*Enter %*	*Cite standard/data*

5. Configuration Artefact Instructions

Once agreed, these values shall be transcribed into the platform configuration file:

```
compliance-config/tolerance-rules.json
```

This file must be:

- Committed to version control with this signed document attached as evidence
- Tagged with the schema version it applies to
- Subject to Compliance Lead sign-off for any future changes

> ⚠ *Any future change to tolerance values constitutes a compliance rule change and must follow the regulatory change management process defined in FR-065 to FR-067.*

6. Sign-off

By signing below, the named individuals confirm that the tolerance values in Section 4 are agreed, defensible against the referenced standards, and approved for use in Phase 1 build.

Role	Name	Signature	Date
Compliance Lead			
Supply Chain Expert			

Role	Name	Signature	Date
CTO (witness)	_____	_____	_____

Once signed, file with project documentation and attach to Jira/Azure DevOps ticket for Decision D3.